

VISUAL UI/UX MOCKUP ADDENDUM

CAPAR Predictive State System

Android Participant App + Web Research Console
Personal Experience Learning + Gamification + EMA 1–4

**SENSE → AWARE → REASON → RECOVER → EXPERIENCE → LEARN →
ENGAGE → ADAPT**

Dokumen tambahan khusus tampilan visual, screen state, dan developer handoff.

Versi visual: Agustus 2026

1. Cakupan Tampilan

Addendum ini memvisualisasikan antarmuka yang sudah didefinisikan pada panduan UI/UX utama. Fokusnya adalah bagaimana state, evidence readiness, baseline journey, episode, EMA, personal experience, dan gamification terlihat pada layar. Semua angka pada mockup adalah data ilustratif dan tidak boleh di-hard-code pada frontend.

Platform	Navigation / screen	Primary purpose	Source binding
Android	Home	Current evidence/state + mission	windows_scored + gamification snapshot
Android	Journey	Evidence maturity, baseline progress, missions, badges	baseline + gamification state
Android	Episode	Episode list/detail + EMA flow	episodes + EMA completion
Android	Insights	Personal experience summaries	personal_experience.json
Android	Profile	Device, consent, notification, privacy	device/consent/preferences
Web	Live Monitor	Near-real-time cohort/participant observability	latest windows + device
Web	Episode	Review trajectory, timing, EMA and outcome	episodes + audit
Web	Experience	Experience memory + gamification observability	experience + gamification
Web	Model & Rules	Active parameters, versions and constraints	config/model versions
Web	Export	Controlled research export	export job + provenance

Visual safety rule: progress, level, badge, and mission represent evidence participation/maturity—not health, fitness, autonomic “quality,” or clinical status.

2. Information Architecture Visual

HOME	JOURNEY	EPISODE	INSIGHTS	PROFILE
state + missions	baseline + level	timeline + EMA	experience	device + consent

Android: EMA 1–4 tidak menjadi tab permanen. EMA muncul sebagai task/modal dari episode atau mission berdasarkan trigger state/event.

LIVE MONITOR	EPISODE	EXPERIENCE	MODEL & RULES	EXPORT
stream + readiness	review + audit	memory + engagement	parameters	dataset/report

Web: cohort selector, participant selector, global date range, search, notification center, dan role badge ditempatkan sebagai konteks global di top bar.

A01. Home Dashboard

Platform
Android

Entry / trigger
App open / foreground

Purpose
Ringkasan evidence readiness, computational state, baseline progress, mission, dan episode aktif.

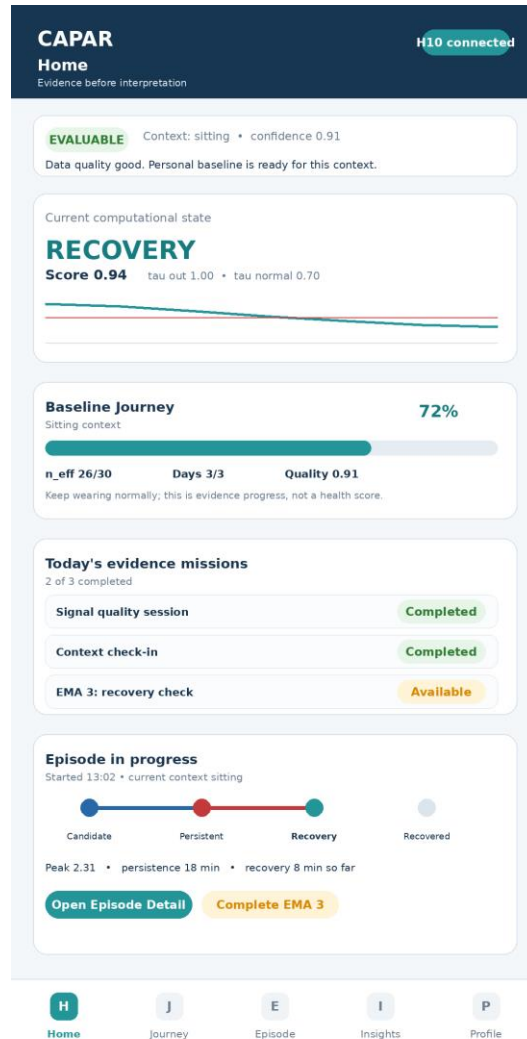


Figure A01 — Home Dashboard. Example values are illustrative.

Key components	Evidence badge; state card; score trajectory; Baseline Journey; today missions; episode progress; bottom navigation.
Primary actions	Open episode detail; complete due EMA; open Journey.
Data binding	evidence_state, physiological_state, anomaly_score, tau_out, tau_normal, baseline progress, gamification missions, episode summary.
UX / safety note	Do not translate EVALUABLE into “healthy”. Progress and missions are evidence-oriented only.

A02. Journey Hub

Platform Android	Entry / trigger Tap Journey	Purpose Menampilkan kematangan evidence, baseline per konteks, mission, dan badge.
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Figure A02 — Journey Hub. Example values are illustrative.

Key components	Evidence Level L1–L6; context baseline journeys; mission center; recent badges.
Primary actions	Open a baseline context; open mission; view all badges.
Data binding	gamification.level, baseline_progress[], missions[], badges[].
UX / safety note	Level must never be described as wellness rank or recovery performance.

A03. Baseline Journey — Context Detail

Platform Android	Entry / trigger Journey → select context	Purpose Menjelaskan apakah baseline personal-context sudah cukup matang untuk inference.
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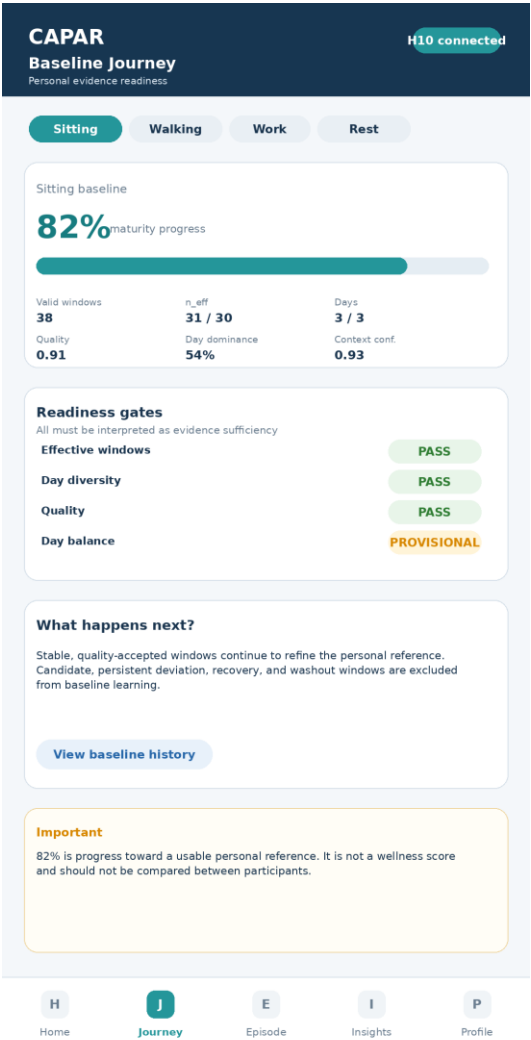


Figure A03 — Baseline Journey — Context Detail. Example values are illustrative.

Key components	Context selector; maturity percentage; n_eff; days; quality; day dominance; readiness gates; explanation.
Primary actions	Switch context; view baseline history.
Data binding	baseline_n, n_eff, min_windows, days, quality, day_dominance, evidence_state.
UX / safety note	If day balance is provisional, explain the limitation rather than silently showing “ready”.

A04. Episode List

Platform Android	Entry / trigger Tap Episode	Purpose Menavigasi episode deviasi–recovery yang auditable.
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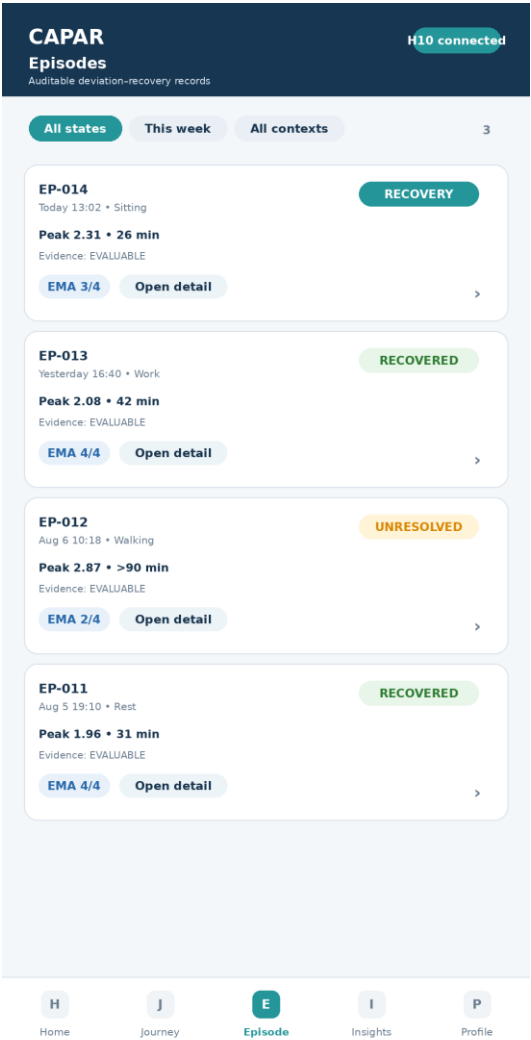


Figure A04 — Episode List. Example values are illustrative.

Key components	Filters; episode cards; state badge; peak/duration; evidence readiness; EMA completion.
Primary actions	Filter; open episode detail; resume pending EMA.
Data binding	episode_id, context, status/state, peak_score, durations, evidence_state, ema_completion.
UX / safety note	Do not sort or reward users by peak score or recovery speed.

A05. Episode Detail & Explainability

Platform
Android

Entry / trigger
Open episode card

Purpose
Melihat trajectory state, threshold, episode evidence, dan alasan transisi.

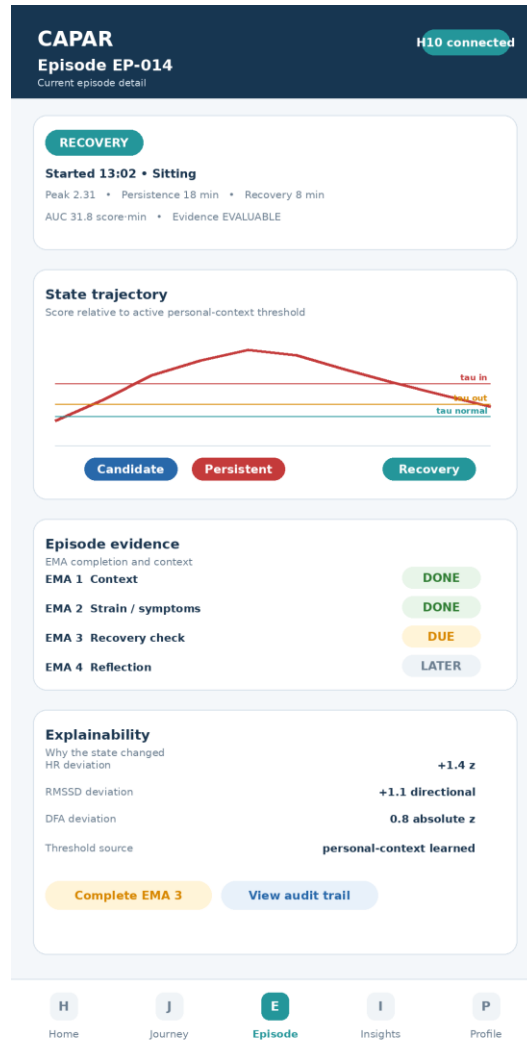


Figure A05 — Episode Detail & Explainability. Example values are illustrative.

Key components	Summary metrics; score trajectory; tau_in/out/normal; state bands; EMA completion; feature deviations; threshold source.
Primary actions	Complete due EMA; view audit trail.
Data binding	episodes.csv + windows_scored + learned threshold source + EMA metadata.
UX / safety note	EMA is annotation/evidence and must not directly rewrite the physiological state in the UI.

A06. Insights — Personal Experience

Platform
Android

Entry / trigger
Tap Insights

Purpose
Menyajikan ringkasan pengalaman recovery personal setelah evidence mencukupi.

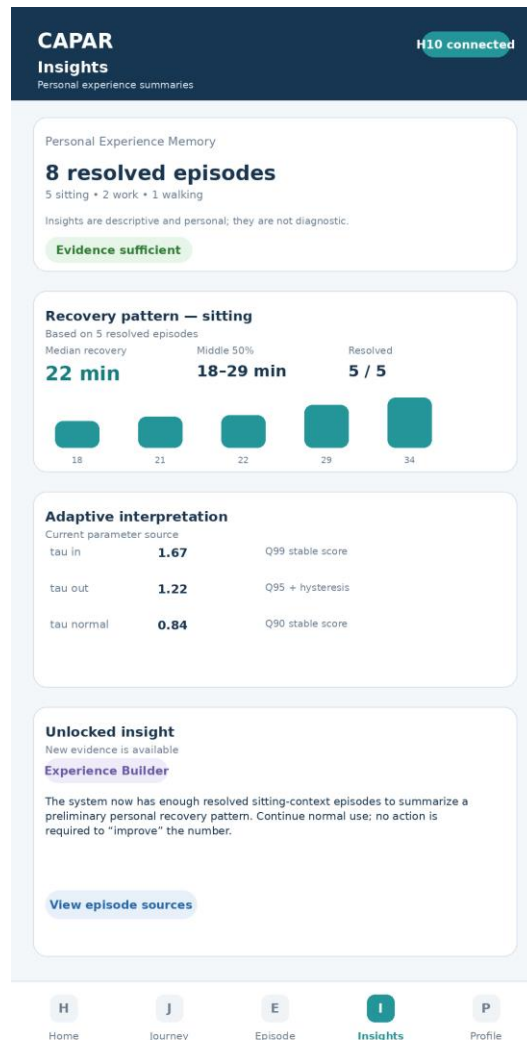


Figure A06 — Insights — Personal Experience. Example values are illustrative.

Key components	Resolved episode count; context distribution; median/IQR recovery; adaptive threshold source; unlocked insight.
Primary actions	Open source episodes; inspect parameter source.
Data binding	personal_experience.json: recovery history, stable-score history, learned thresholds.
UX / safety note	Use “personal pattern” / “preliminary insight”; avoid diagnosis or normative comparison against other users.

A07. Badges & Evidence Levels

Platform Android	Entry / trigger Journey → Badges	Purpose Menampilkan gamification yang berorientasi partisipasi dan kualitas evidence.
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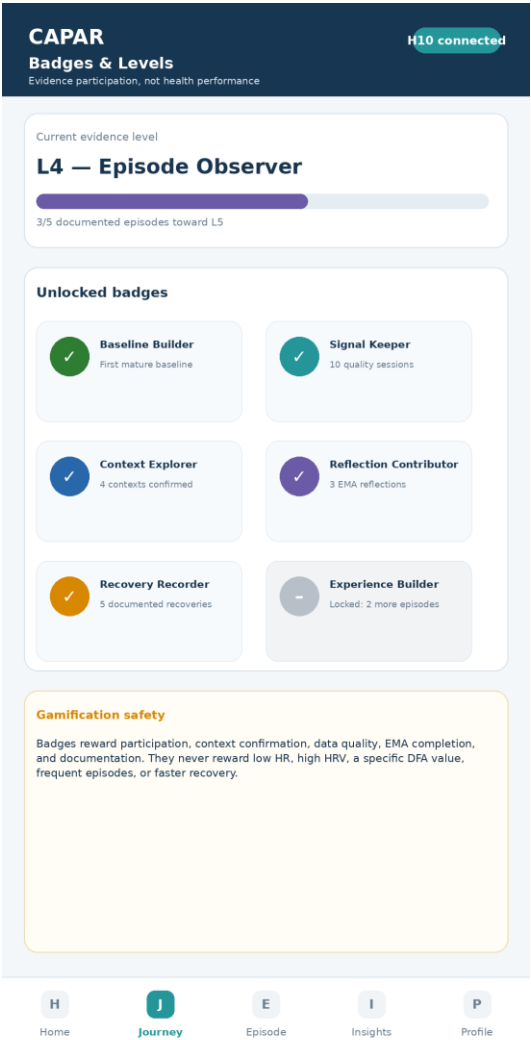


Figure A07 — Badges & Evidence Levels. Example values are illustrative.

Key components	Current level; progress to next evidence level; badge cards; safety message.
Primary actions	Open badge details; inspect requirements.
Data binding	gamification_snapshot.level, badges, progress counters.
UX / safety note	No reward from HR, RMSSD, DFA, anomaly score, episode frequency, or fast recovery.

A08. Profile, Device & Privacy

Platform Android	Entry / trigger Tap Profile	Purpose Kontrol device, consent, notification, quiet hours, dan privacy.
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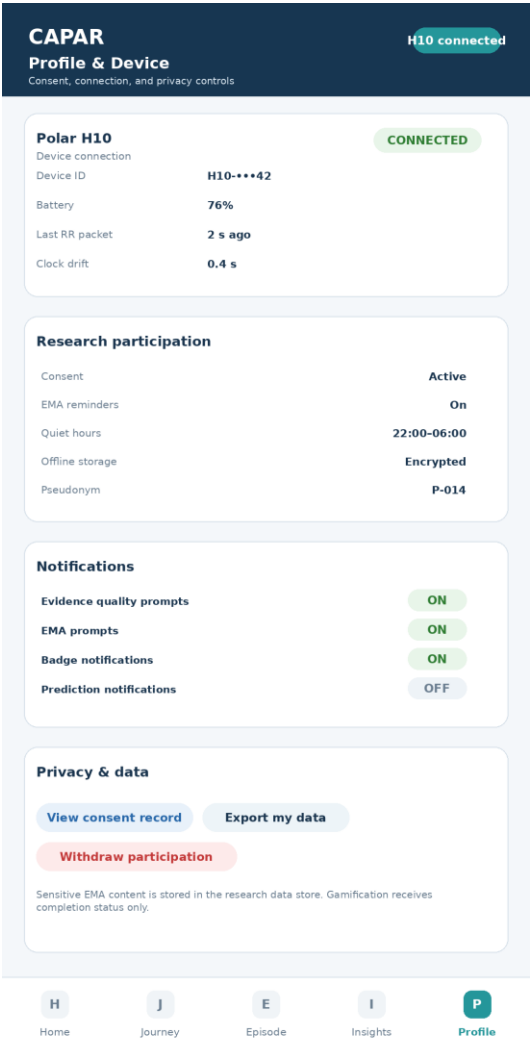


Figure A08 — Profile, Device & Privacy. Example values are illustrative.

Key components	Polar H10 status; battery; sync/clock; consent; notifications; data/export; withdrawal.
Primary actions	View consent; export own data; change notification preferences; withdraw.
Data binding	device status, consent state, notification config, local/offline sync status.
UX / safety note	Withdrawal requires explicit confirmation; sensitive EMA content is not copied into gamification storage.

A09. EMA 1 — Context Confirmation

Platform Android	Entry / trigger DEVIATION_CANDIDATE or UNCERTAIN_CONTEXT	Purpose Mengonfirmasi aktivitas dan konteks pada onset/candidate.
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13:07

EMA 1 — Konfirmasi Konteks

Muncul saat candidate atau konteks tidak yakin

1 2 3 4

Apa aktivitas Anda saat ini?

☒ Duduk / istirahat

☐ Berjalan

☐ Bekerja / belajar

☐ Aktivitas fisik

Apakah aktivitas ini direncanakan?

☒ Ya

☐ Tidak

Catatan opsional

Mis. baru selesai berjalan...

Kirim EMA

Nanti ingatkan saya • Lewati dengan alasan

Figure A09 — EMA 1 — Context Confirmation. Example values are illustrative.

Key components	Progress 1/4; activity choice; planned/unplanned; optional note; submit/snooze/skip.
Primary actions	Submit; remind later; skip with reason.
Data binding	episode_id, trigger_type, trigger_time, response_time, context_confirmed, activity, completion_status.
UX / safety note	Quick choices first. EMA is not a diagnostic classifier.

A10. EMA 2 — Strain / Symptom Check

Platform
Android

Entry / trigger
PERSISTENT_DEVIATION

Purpose
Mencatat pengalaman subjektif ketika deviasi persisten.

Figure A10 — EMA 2 — Strain / Symptom Check. Example values are illustrative.

Key components	Progress 2/4; perceived strain; symptom chips; context note; submit/snooze/skip.
Primary actions	Submit; remind later; skip.
Data binding	episode_id, perceived_strain, symptom flags, response latency, completion status.
UX / safety note	Symptoms are participant-reported research evidence, not automated clinical interpretation.

A11. EMA 3 — Recovery Check

Platform
Android

Entry / trigger
RECOVERY

Purpose
Mengonfirmasi persepsi pengguna ketika sistem mulai recovery.

Figure A11 — EMA 3 — Recovery Check. Example values are illustrative.

Key components	Progress 3/4; feel-back-to-usual scale; activity/context confirmation; submit/snooze/skip.
Primary actions	Submit; remind later; skip.
Data binding	episode_id, perceived_recovery, context, completion status, latency.
UX / safety note	Do not ask user to force a faster recovery or hit a target physiological value.

A12. EMA 4 — Post-Episode Reflection

Platform Android	Entry / trigger RECOVERED / episode close	Purpose Refleksi pascaepisode untuk experience memory dan review.
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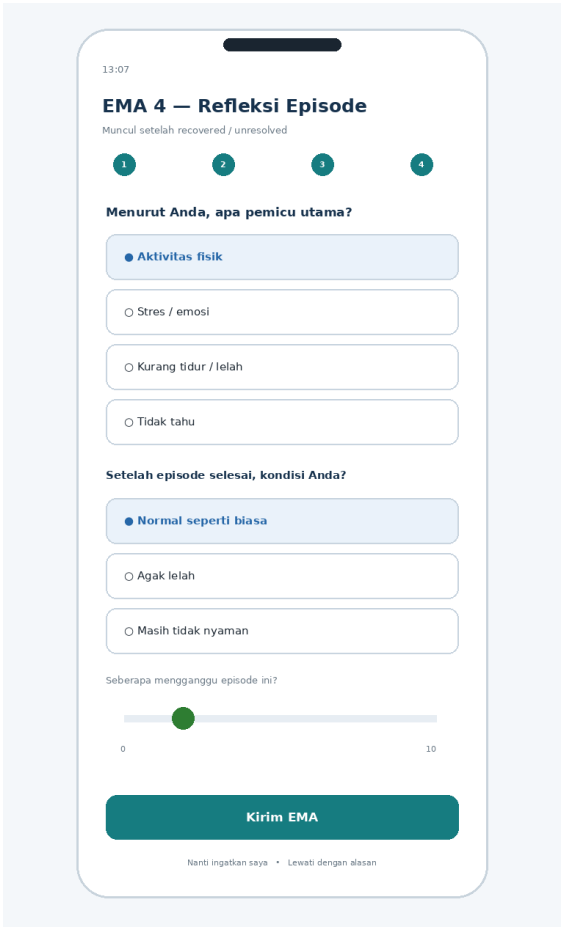


Figure A12 — EMA 4 — Post-Episode Reflection. Example values are illustrative.

Key components	Progress 4/4; likely trigger/context; optional reflection; perceived resolution.
Primary actions	Submit reflection; skip with reason.
Data binding	episode_id, reflection tags, perceived trigger, resolution, completion metadata.
UX / safety note	Reflection can inform later review/recalibration only through governed workflow.

W01. Live Monitor

Platform
Web Research Console

Entry / trigger
Sidebar → Live Monitor

Purpose
Near-real-time observability pada participant dan cohort.

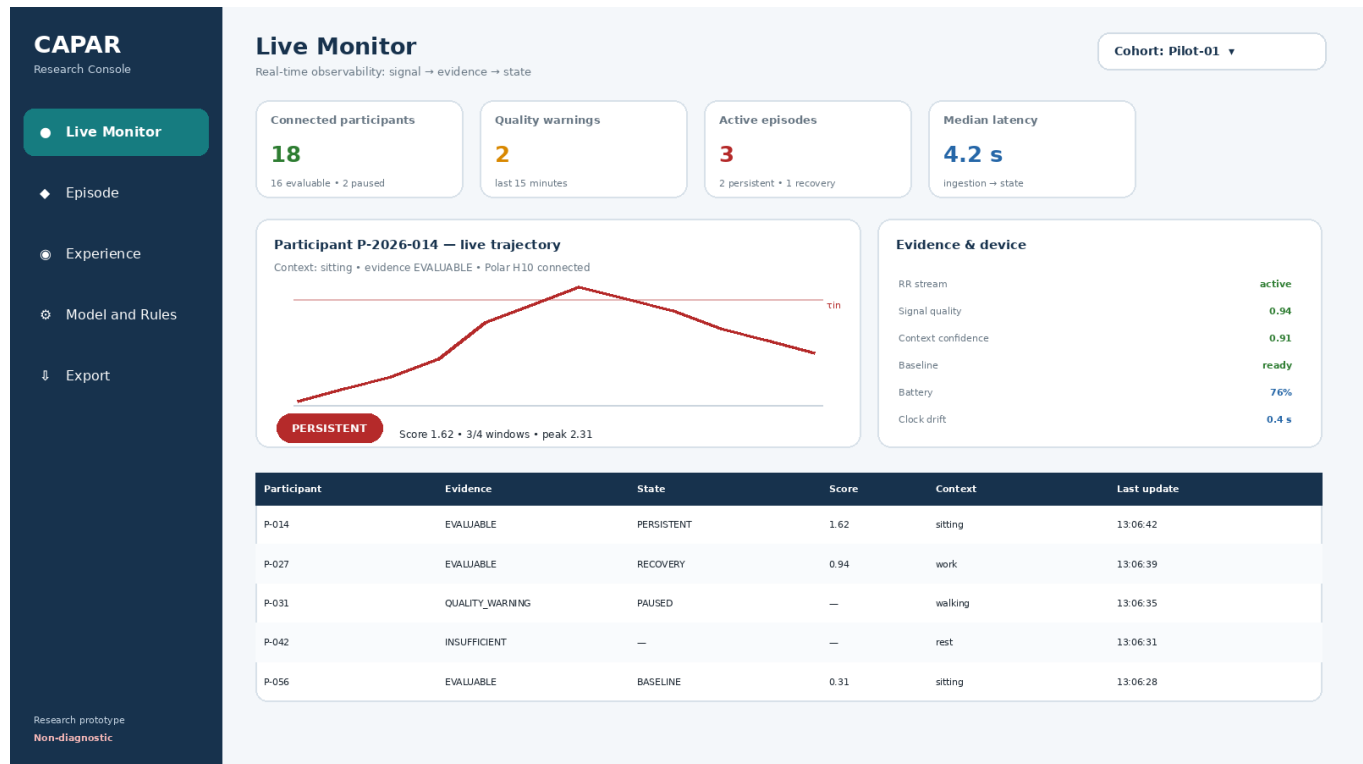
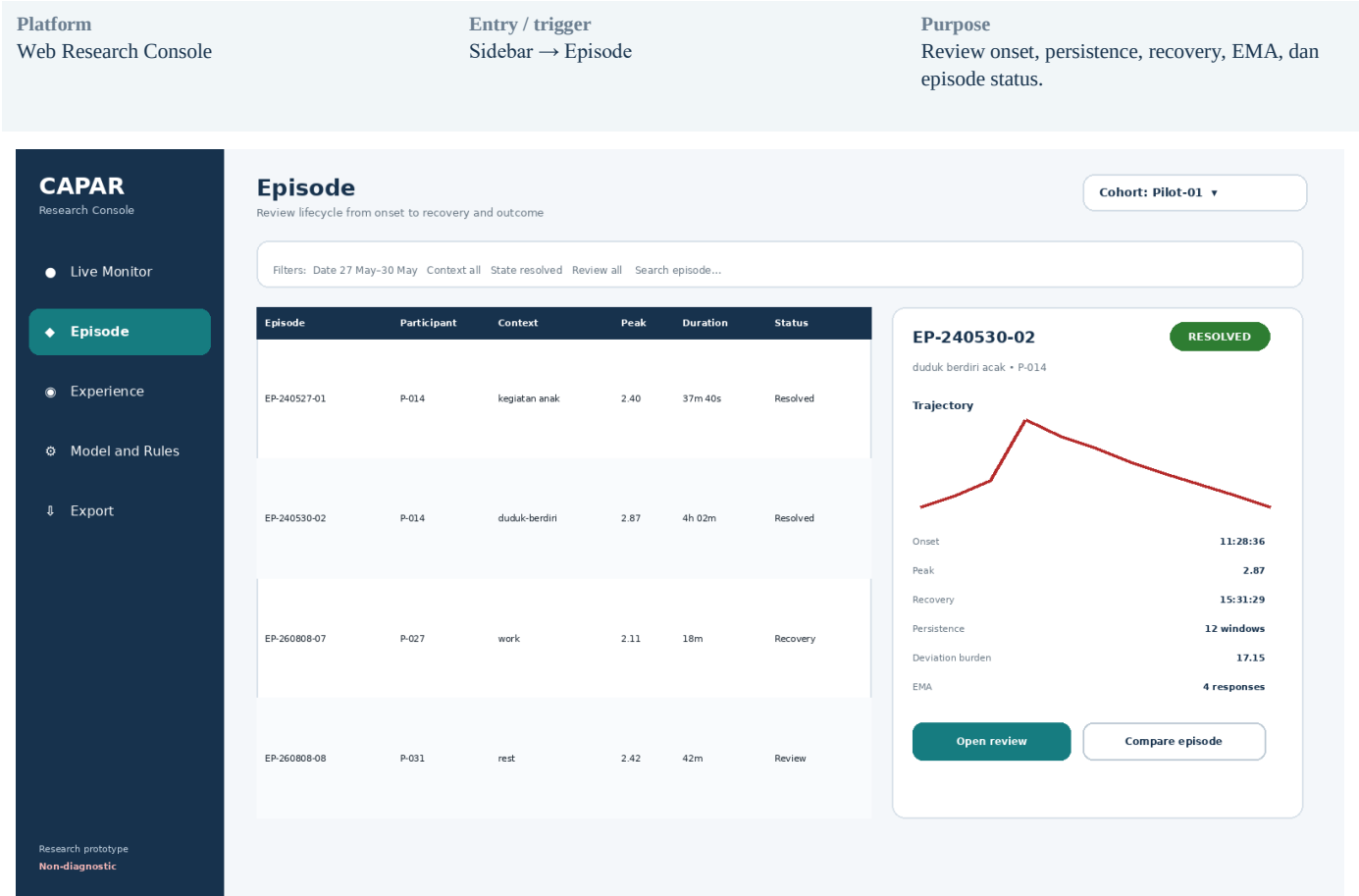


Figure W01 — Live Monitor. Example values are illustrative.

Key components	KPI cards; live trajectory; evidence/device panel; participant table; active alerts.
Primary actions	Filter participant/date; flag session; open episode.
Data binding	latest windows, evidence_state, physiological_state, anomaly_score, context, device quality/battery.
UX / safety note	Evidence warnings should suppress over-interpretation and show why inference is unavailable.

W02. Episode Review



W03. Experience

Platform Web Research Console	Entry / trigger Sidebar → Experience	Purpose Observability Personal Experience Memory dan adaptation evidence.
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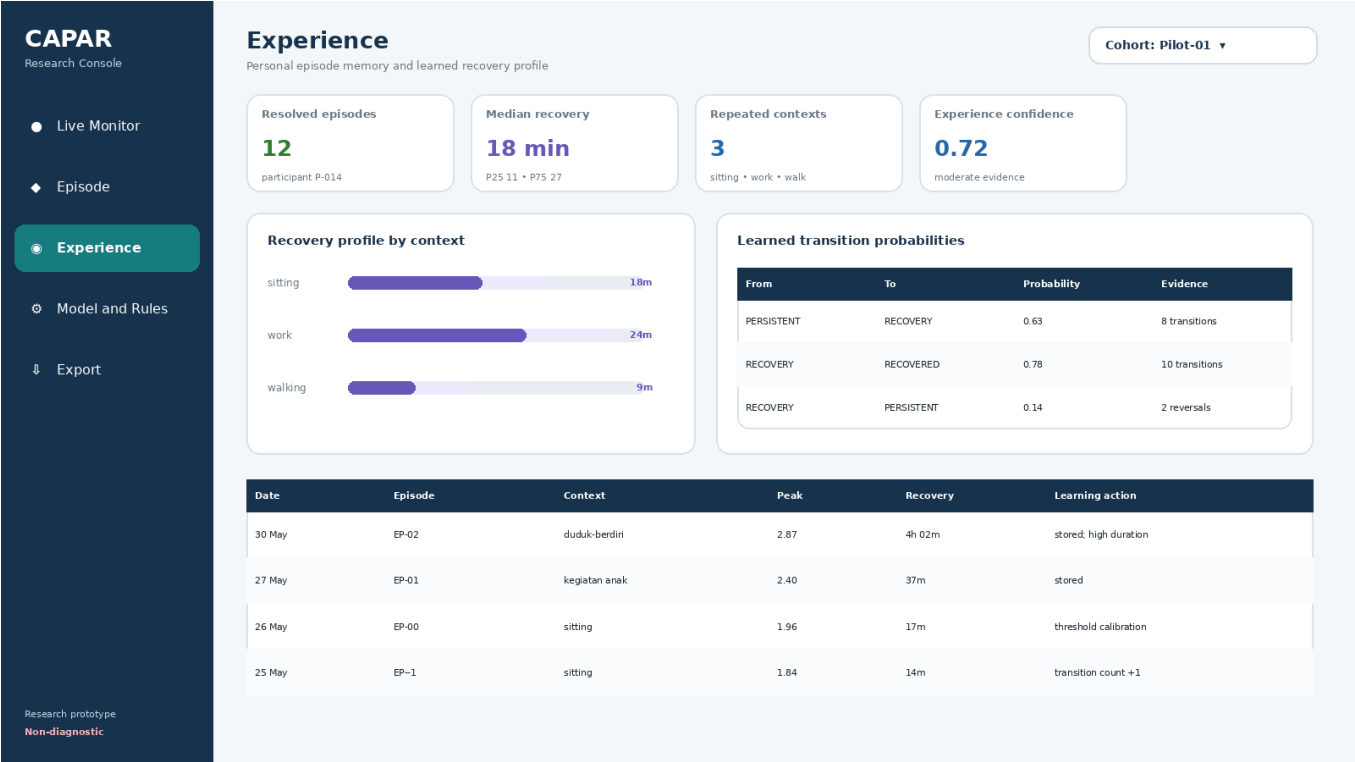
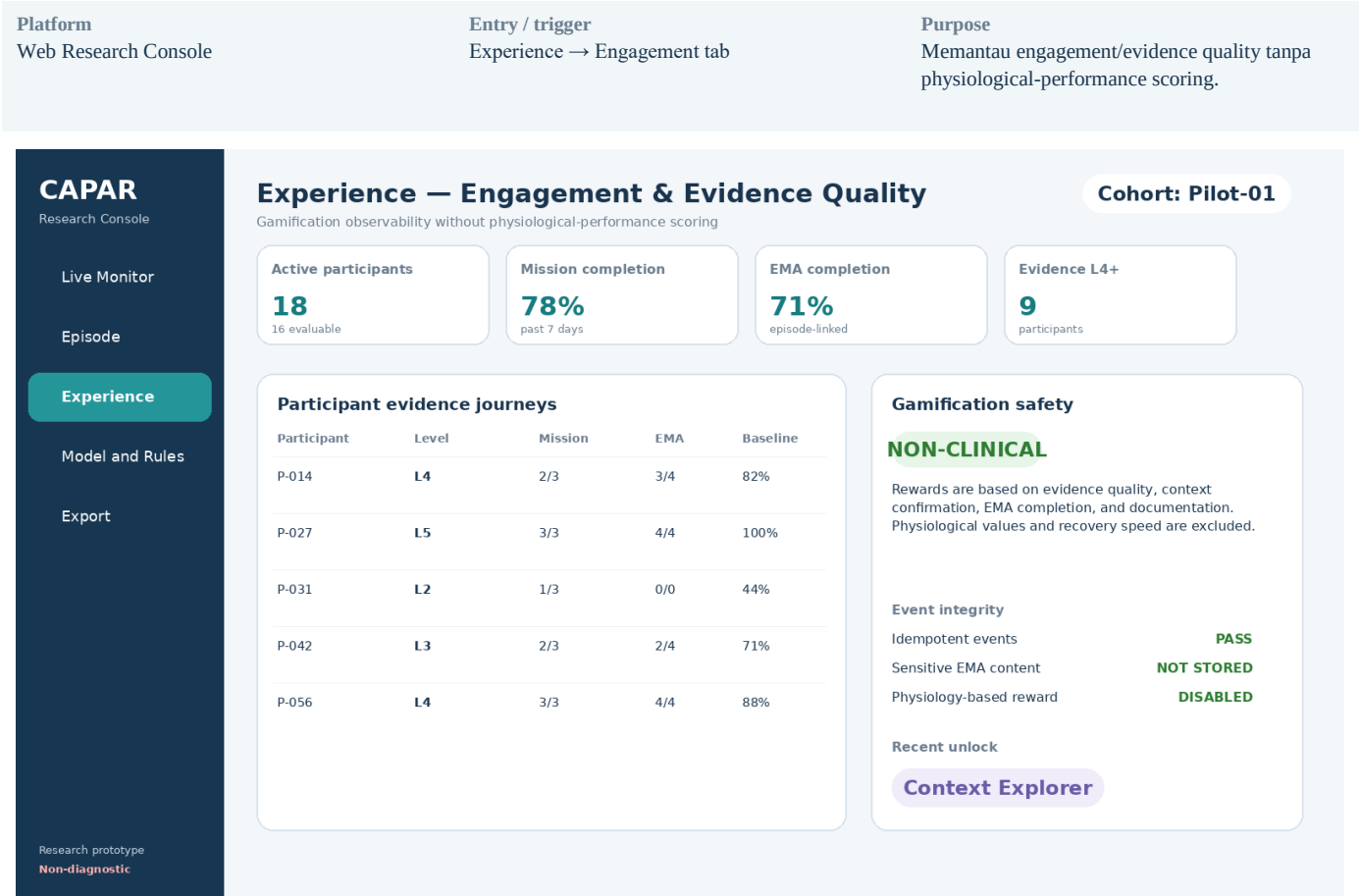


Figure W03 — Experience. Example values are illustrative.

Key components	Resolved episode summaries; recovery distribution; learned thresholds; experience maturity.
Primary actions	Filter user/context; open source episodes; compare parameter versions.
Data binding	personal_experience.json, stable_score history, transition/recovery memory.
UX / safety note	Show sample size and uncertainty; never present learned values as clinical reference ranges.

W04. Experience — Gamification Observability



W05. Model & Rules

Platform Web Research Console	Entry / trigger Sidebar → Model and Rules	Purpose Menampilkan parameter aktif, versioning, threshold source, dan governance.
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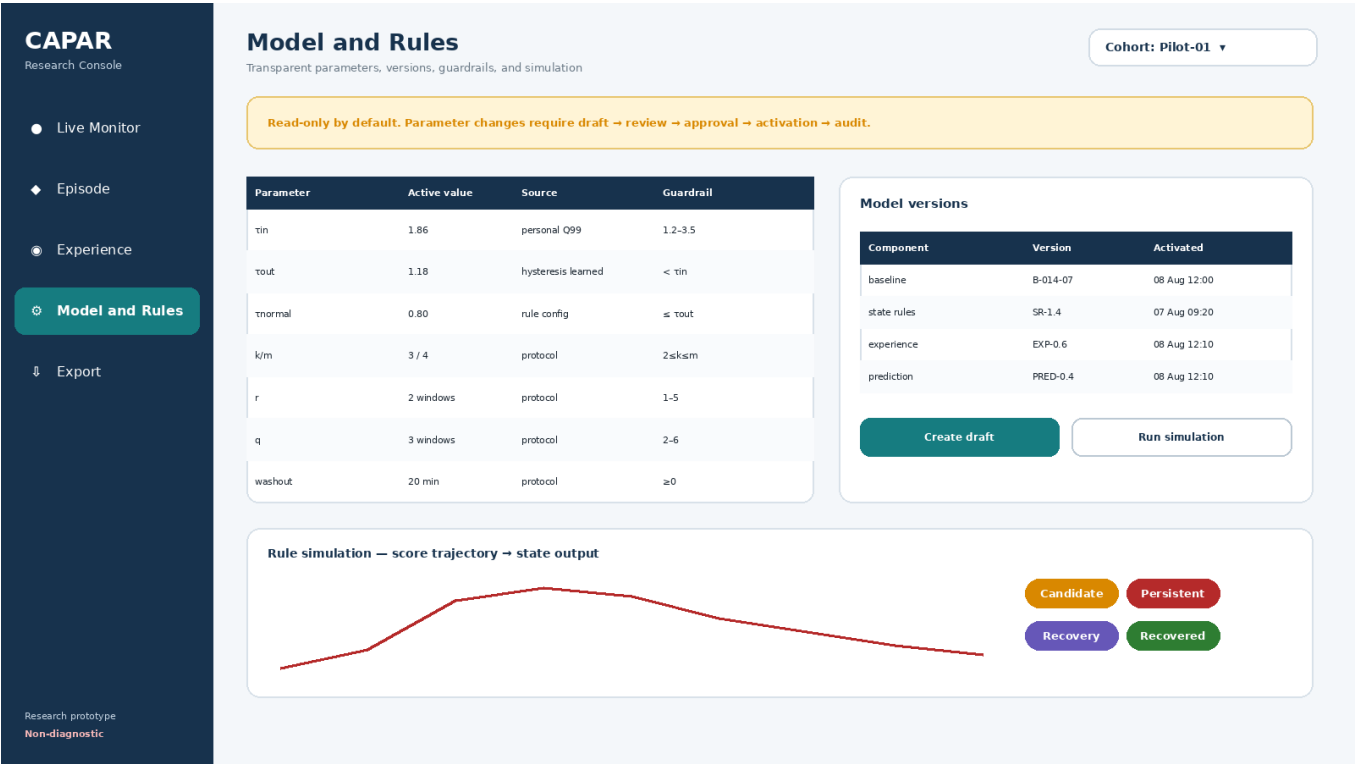


Figure W05 — Model & Rules. Example values are illustrative.

Key components	Active config; thresholds; state rules; model version; audit/change history.
Primary actions	Compare version; stage change; export config; reviewer read-only mode.
Data binding	config JSON, model/rules version, threshold source, calibration metadata.
UX / safety note	UI must read parameters from engine/config; never hard-code tau or weights.

W06. Export

Platform Web Research Console	Entry / trigger Sidebar → Export	Purpose Membangun paket ekspor terkontrol untuk penelitian dan review.
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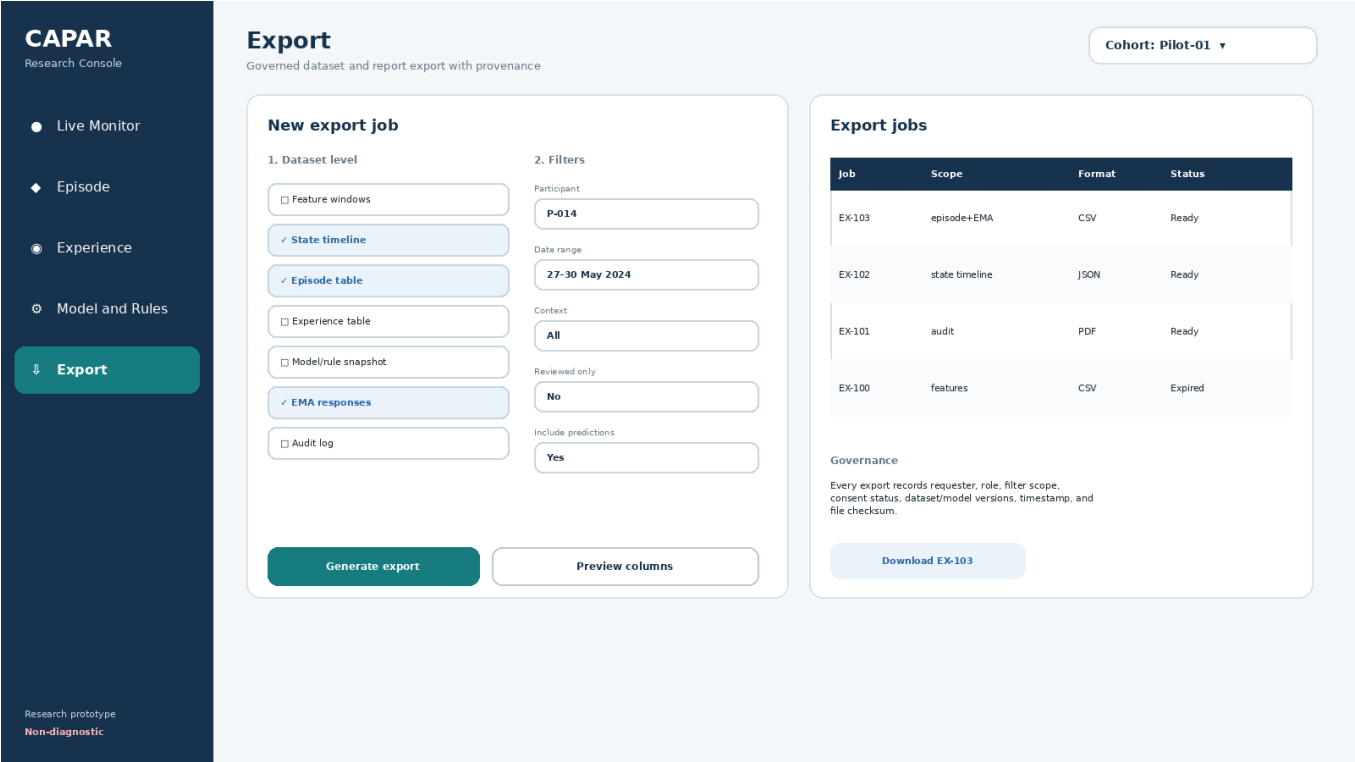


Figure W06 — Export. Example values are illustrative.

Key components	Scope; date/context filters; data selection; pseudonymization; format; provenance.
Primary actions	Create export job; download when complete; inspect manifest.
Data binding	windows, episodes, EMA metadata, experience, gamification summary, config/version provenance.
UX / safety note	Default to pseudonymized export; raw identifiers require explicit role/permission.

3. Developer Handoff — Visual Acceptance Checklist

- ☐ Android bottom navigation konsisten: Home, Journey, Episode, Insights, Profile.
- ☐ EMA 1–4 muncul sebagai task/modal berbasis trigger, bukan sebagai tab permanen.
- ☐ Setiap layar yang menampilkan physiological state juga menampilkan evidence readiness atau link ke evidence detail.
- ☐ Threshold, weight, state, score, prediction, dan experience values berasal dari engine/config aktif.
- ☐ Baseline progress dan evidence level tidak diberi label “health score”.
- ☐ Badge/mission tidak boleh menggunakan HR, RMSSD, DFA, anomaly score, episode frequency, atau recovery speed sebagai reward.
- ☐ Offline EMA/gamification event bersifat idempotent saat resync.
- ☐ Sensitive EMA answer content tetap pada research data store; gamification hanya menerima completion/event metadata.
- ☐ Web menu inti tersedia: Live Monitor, Episode, Experience, Model & Rules, Export.
- ☐ Export menyertakan provenance: participant pseudonym, time scope, data version, model/config version, dan export job ID.

Prototype status: mockup visual ini adalah design-document evidence. Usability, runtime feasibility, dan clinical effectiveness harus dibuktikan melalui implementasi dan pengujian terpisah.